

Wenhe (Zachary) Zhao

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EDUCATION

Honors Bachelor of Science in Engineering (Grad May 2027)

The Ohio State University – Columbus, OH

Cumulative GPA: 3.79

Major: Engineering Physics

Mechanical Engineering Track

Honor Student

Honor Research Distinction

RESEARCH EXPERIENCE

Nanoengineering and Biodesign Laboratory – The Ohio State University

Columbus, OH

Advisor: Prof. Carlos Castro

April 2024 – Present

Research as part of NSF DMREF project focusing on “Programmable Dynamic Actuation of DNA Origami Devices and Assemblies”

Projects on Experiment:

Programmable Long-range Rolling Motion of DNA Origami Nano-assemblies

- Design and folded the gear-track DNA origami nanomotor, including scaffold routing, staple layouts, and sequence programmed sticky-end interfaces for dynamic actuation.
- Established on-track binding/stepping experiment, imaged step states and orientation via TEM.

Dynamics Assemblies from Multi-arm DNA Devices with Tunable Mechanical Properties

- Folded tri-arm variants with tunable hinge stiffness and arm geometry.
- Purified assemblies (AGE/PEG) and validated structures by TEM.
- Exploring mechanical response by varying sticky-end interfaces and linker designs; compared trends with simulations.

Biological & Soft Materials Modeling Lab -- Duke University

Durham, NC

Advisor: Prof. Gaurav Arya

May 2025 – Present

Develop and simulate the “Programmable Dynamic Actuation of DNA Origami Devices and Assemblies”

Projects on Simulation:

Programmable Long-range Rolling Motion of DNA Origami Nano-assemblies

- Built statistical-mechanics models to map sequence dependent free-energy landscapes and transition rates.
- Simulated sequence-coded gear-track interfaces to study directional, stepwise motion and processivity.

Dynamics Assemblies from Multi-arm DNA Devices with Tunable Mechanical Properties

- Built coarse-grained oxDNA/LAMMPS Mesoscopic models of a multi-arm device with tunable hinge stiffness and arm geometry.
- Quantified elastic response and inter-arm coupling; generated energy-deformation and state population curves.

BatheBioNano Lab -- Massachusetts Institute of Technology (MIT)

Boston, MA

Advisor: Prof. Mark Bathe

June 2026 – Present

Project: Develop and simulate the “pH-Controlled Enzyme Functionalization on Chip”

- Designed i-motif-based pH-switching experiments and planned AFM/fluorescence readouts to evaluate enzyme binding.
- Screened enzyme candidates for pH-dependent functionalization/de-functionalization on DNA origami and chip platforms.
- Developed coarse-grained enzyme-DNA origami models to support experimental design.
- Performed all-atom simulations of enzyme silicification and DBCO-DNA functionalization.

Machining Research Laboratory – CENTECH-EG

Sichuan, China

Mechanical Engineering Research

May 2023 – August 2023

Tool Life Prediction for Turning Inserts

- Collaborated with the team to analyze measurement data, offering valuable suggestions for improvement.
- Contributed to a key R&D project: "Development and Application of High Precision, High Efficiency, High-Speed Multi-Facet Intelligent Turning Tool System" in Sichuan Province.

INDUSTRY EXPERIENCE

CENTECH-EG

Sichuan, China

Mechanical Engineering Intern (TOP 1 high end tool company in province)

May 2023 – August 2023

- Developed proficiency in using NX modeling software.
- Operated HASS vertical and horizontal machining centers, and the Mazak CNC turning machine for independent parts machining.
- Learned specific operations of Alicona microscopes, blade fundamentals, and related knowledge.

PUBLICATIONS

Published Paper

- [1] Lei,H; Zhao,W., et al. (2024). **Optimizing Tool Wear Prediction in Intelligent Manufacturing: A Multi-Sensor Approach Enhanced by RealNVP.** *Measurement Science and Technology*. <https://doi.org/10.1088/1361-6501/ad77ea>

Ongoing Projects in Preparation

- [2] [†]Huang,K; [†]Zhao,W.; Duke,D., et al. **Programmable Long-range Rolling Motion of DNA Origami Nano-assemblies** (Plan to submit on *Nature Nanotechnology* 2026 Fall)
- [3] [†]Zhao,W.; [†]Ma,S; Duke,D., et al. **Understanding Programmable Gear-Track DNA Origami Systems via Multiscale Coarse-Graining Modeling and Experiments** (Plan to submit on *ACS nano* 2027 Spring)
- [4] [†]Zhao,W.; [†]Ma,S., et al. **Characterizing DNA Origami Nanostructures in TEM Images via Molecular Dynamic Assisted Machine Learning** (Plan to submit on *Advance Science* 2026 Fall)
- [5] W. Zhao, S. Ma., **Undersanding Reverse Design of i-motif pH Swith** (Plan to submit on *JCTC* 2026 Fall)
- [6] Teng,T; Duke, D; Zhao,W., et al. **Dynamics Assemblies from Multi-arm DNA Devices with Tunable Mechanical Properties**
- [7] Xu, H; Teng, T; Zhao, W., et al. **Controlling DNA origami with polynucleotide brushes**

[†]Denotes Planned equal (co-first) authorship

CONFERENCE PRESENTATIONS

Materials Research Society (MRS) Fall Meeting — Poster Presenter,

Boston, MA, Dec 2025

- [1] [†]Zhao,W.; [†]Huang,K; Pfeifer,W., et al. “Programmable Processive Motion of DNA Origami Nano-assemblies”
— The Best Poster Award Nominee
- [2] Teng,T; Duke,D; Zhao,W., et al. “Probing Mechanical Properties and Dynamic Assembly of DNA Origami by Coarse-grained Molecular Dynamics Simulations”

31st Annual Denman Undergraduate Research Forum — Poster Presenter,

Columbus, OH, Mar 2026

- [3] Zhao,W.; Duke,D; Huang,K., et al. “Mesoscopic Simulation of Dynamic DNA Origami Nanomachine” — First Place Award

SVC 19th Semi-Annual Spring 2026 Meeting — Poster Presenter,

Columbus, OH, May 2026

- [4] [†]Zhao,W.; [†]Huang,K; Duke,D., et al. “Dynamic Interactions of a Nanoscale Rack and Pinion Assembly”
- [5] Teng,T; Duke,D; Zhao,W., et al. et al. “A Simulation Guide Design of Multi-arm Nanostructures for Higher Order Architected Material Assembly”

2026 Institute for Materials and Manufacturing Research (IMR) Conference — Poster Presenter,

Columbus, OH, May 2026

- [6] Zhao,W.; Duke,D., et al. “A Free Energy Guided Mesoscopic Model for Large-Scale Dynamical DNA Origami Systems”
- [7] [†]Zhao,W.; [†]Huang,K; Duke,D., et al. “Dynamic Interactions of Nanoscale Rack and Pinion Assembly”

oxDNA Conference and Workshop — Poster Presenter,

Tempe, AZ, Feb 2026

- [8] [†]Zhao,W.; [†]Huang,K; Pfeifer,W., et al. “Coarse-grained Modeling of Rolling-based DNA Origami Nanomotor”
- [9] Teng,T; Duke,D; Zhao,W., et al. “Mesoscopic Dynamics Simulations of DNA Origami Mechanical Response and Assembly Behavior”

MAE Buckeye Engineering Grad Program Research Showcase — Poster Presenter,

Columbus, OH, Oct 2025

[10] [†]**Zhao,W;** [†]**Huang,K;** Pfeifer,W., et al. “DNA Origami Nanomachines”

MidWest Nanotech (miDNAno) Workshop — Oral Talk

Columbus, OH, Jul 2026

[11] [†]Ma,S; [†]**Zhao,W.**, “Characterizing DNA Origami Nanostructures in TEM Images via Molecular Dynamic Assisted Machine Learning”

MidWest Nanotech (miDNAno) Workshop — Oral Talk

Columbus, OH, Jul 2026

[11] **Zhao,W.**, “Dynamic of Programmable DNA Origami Motors”

PATENT

Universal Floating Tool Holder — Zhao.W

Issued Aug 4, 2023 - ZL 2023 2 0418328.X

- **Elastic Support Ring:** Introduced a support ring in the rear clamping cavity of the tool holder's main body to provide resetting force in all directions, enhancing grinding quality.
- **Articulated Bearing Placement:** Implemented an articulated bearing in the front clamping cavity to centralize the oscillation pivot, reducing amplitude and improving grinding precision.

Nucleic Acid Test Sampling Tube — Zhao.W

Issued Jul 5, 2023 - ZL 2022 2 0891815.3

- Providing a nucleic acid test sampling tube with a new cover mechanism, improving the efficiency of opening and closing procedures, that facilitated quicker and more convenient usage, contributing to the efficiency of COVID-19 testing procedures.

AWARDS

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| ▪ 1 st Place Award in Engineering at 2026 Denman Research Forum | - \$500 |
| ▪ 2025 College of Engineering Scholarship (Balley Endowment Fund) | - \$2000 |
| ▪ 2026 Undergraduate Research Scholarship | - \$2000 |
| ▪ 2026 Perry Research Scholarship | - \$2000 |
| ▪ 2026 Academic Enrichment Grants | - \$1000 |
| ▪ MRS The Best Poster Nominee (2025) | |
| ▪ Honor Research Distinction (2026) | |
| ▪ College of Engineering Dean's List | |

SKILLS

- **Experimental Skills:** Pipetting, Agarose Gel Electrophoresis, Agarose Gel Extraction, DNA nanostructure self-assembly and purification, Nanodrop, Buffer Preparation, Fluorescence Readout, Transmission Electron Microscopy (TEM), Atomic Force Microscopy (AFM)
- **Simulation & Design Tools:** oxDNA, LAMMPS, GROMACS, caDNAno, MagicDNA, ANSYS, CAD, Onshape, Solidworks
- **Programming Languages:** Python, MATLAB, Mathematica, CNC programming
- **Other Research Skills:** Molecular Dynamic Simulation, Machine Learning, Statistical Mechanics, Monte Carlo Simulation, Experimental Design, Data Analysis, Hypothesis Testing, Statistical Modeling, Paper Writing, Literature Review

REFERENCES

Dr. Carlos E. Castro

Professor of Mechanical and Aerospace Engineering at the Ohio State University, **Email:** castro.39@osu.edu

Dr. Gaurav Arya

Professor of Biomedical Engineering at the Duke Engineering, **Email:** gaurav.arya@duke.edu

Dr. Mark Bathe

Professor of Biological Engineering at the Massachusetts Institute of Technology (MIT), **Email:** mbathe@mit.edu